Data Science - ING 3 BI

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***Abstract*—This project explores how beer characteristics influence their overall rating using machine learning techniques. A publicly available dataset containing various beer attributes (such as bitterness, body, fruity or sour notes) was analyzed and preprocessed to remove irrelevant and potentially biasing information. A Random Forest Regressor was implemented to predict the overall rating of beers based on their intrinsic features. The model achieved moderate performance, highlighting the complexity of human taste preferences. In addition to predictive modeling, exploratory data analysis was conducted to identify key differences between highly rated and poorly rated beers. Visualization techniques such as feature importance plots and heatmaps were used to better understand the impact of specific attributes and their combinations. Finally, synthetic beer profiles were generated to identify optimal combinations of characteristics that maximize predicted ratings. This approach provides insights into what defines a well-balanced and highly appreciated beer.**

***Index Terms*—beer analysis, machine learning, data science**

I. Introduction

Understanding what makes a beer highly rated is a complex problem, as it depends on multiple interacting characteristics such as bitterness, sweetness, alcohol content, and aromatic profiles. With the increasing availability of structured datasets, machine learning offers an effective way to analyze these factors and uncover hidden patterns. The objective of this project is to predict the overall rating of a beer based solely on its intrinsic properties, without relying on subjective review components. This allows for a more objective analysis of how specific features influence user perception. To achieve this, a dataset containing detailed beer profiles and ratings was used. After data cleaning and preprocessing, a Random Forest regression model was trained to learn the relationship between beer attributes and their ratings. The model's performance was evaluated using standard metrics such as Mean Squared Error (MSE) and R^2 score. Beyond prediction, the project also focuses on interpretability. By comparing top-rated and low-rated beers, and analyzing feature importance, it is possible to identify the key characteristics that contribute to a high-quality beer. Additional visualizations, including heatmaps and synthetic profile simulations, were used to explore optimal combinations of features. This project demonstrates how machine learning can be applied to a real-world problem in the food and beverage domain, providing both predictive capabilities and actionable insights.

II. Problem Definition

For this project, we aim to determine which factors in a beer have the greatest influence on consumer ratings. Beer evaluation is a complex process driven by multiple factors such as bitterness, sweetness, alcohol content, body, and aromatic notes (e.g., fruity or sour profiles).

To achieve this, we want to predict how these different characteristics interact with each other and influence the overall rating.

We are therefore dealing with a regression problem, where a global rating is influenced by these various factors. Our objective is to predict the overall rating of a beer using its intrinsic characteristics, identify the most influential features driving beer quality, and explore combinations of features that lead to highly rated beers.

III. Dataset Description

The dataset used in this project is the *Beer Profile and Ratings* dataset, which contains detailed information about beers, including their chemical properties, sensory characteristics, and user reviews.

Originally, the dataset includes a wide range of features:

- **Identification and textual data:** *Name, Beer Name (Full), Brewery, Description*
- **Chemical and physical properties:** *ABV, Min IBU, Max IBU*
- **Sensory attributes:** *Astringency, Body, Alcohol, Bitter, Sweet, Sour, Salty, Fruits, Hoppy, Spices, Malty*
- **User review scores:** *review_aroma, review_appearance, review_palate, review_taste, review_overall*
- **Other information:** *number_of_reviews*

However, several columns were removed during preprocessing. Textual features such as beer names, brewery, and descriptions were excluded, as they are not directly usable in numerical modeling without advanced natural language processing. Additionally, some review sub-scores were removed to avoid redundancy with the overall rating.

IV. Exploratory Data Analysis

To explore the dataset, we used Power BI to create various visualizations focusing on individual features.

The objective was to analyze how each characteristic (such as bitterness, sweetness, or fruity notes etc..) influences the overall beer rating. This allowed us to identify potential trends and relationships between features and perceived quality.

Additionally, we examined the distribution of ratings to ensure that the dataset was well-balanced and suitable for machine learning. This step was important to verify that the model would not be biased toward a specific range of ratings.

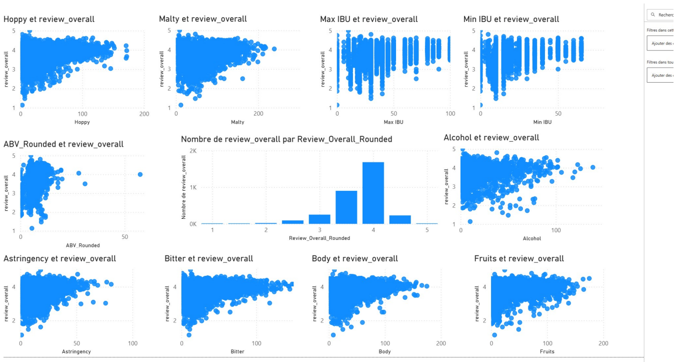

Fig. 1. Relationship Between Beer Features and Overall Rating

This figure presents multiple scatter plots showing the relationship between different beer characteristics and the overall rating.

Overall, most features such as Hoppy, Malty, Body, Bitter, and Fruits show a slight positive correlation with the rating. Beers with higher values in these attributes tend to achieve better scores, although the relationship remains weak and highly dispersed.

On the other hand, features like IBU (both Min and Max) and ABV do not show a clear linear relationship with the rating, suggesting that extreme bitterness or alcohol content alone does not guarantee a higher appreciation.

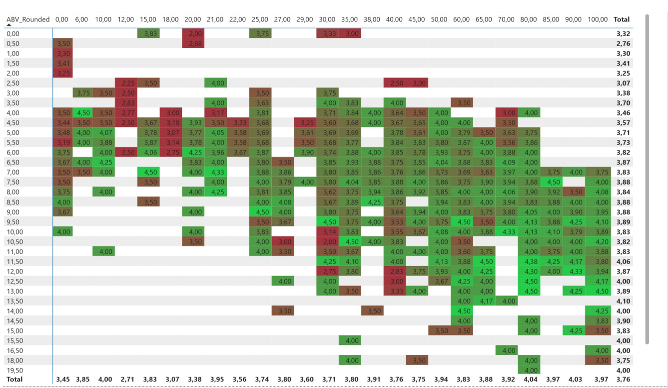

Fig. 2. Relationship Between ABV and Max IBU related to Overall rating

For this second figure, we attempted to combine two factors and observe their impact on the ratings. We therefore created a matrix with ABV on the rows and MAX IBU on the columns.

This observation highlights that beer quality is not determined by a single factor, but rather by a combination of characteristics, justifying the use of a machine learning model to capture these complex interactions.

V. Methodology

The methodology of this project is based on a structured machine learning pipeline, combining data preprocessing, model training, and result interpretation.

First, the dataset was cleaned by removing non-relevant and textual features such as beer names, descriptions, and detailed review components (aroma, appearance, taste, etc.). This step was essential to avoid data leakage and ensure that the model relies only on intrinsic beer characteristics.

The remaining features were then separated into input variables (X) and the target variable (y), which corresponds to the overall beer rating. Since the dataset includes a categorical feature (Style), it was encoded using a LabelEncoder to make it compatible with machine learning algorithms.

The dataset was split into training and testing sets using an 80/20 ratio. This allows the model to learn from the majority of the data while keeping a portion unseen for evaluation, ensuring a reliable assessment of its generalization ability.

A Random Forest Regressor was chosen as the predictive model. This algorithm is particularly suitable for this problem because it can capture non-linear relationships and interactions between features, which are expected in taste-related data. The number of estimators was set to 100 as a compromise between performance and computational efficiency. Indeed fewer than that would not be precise enough, and more does not significantly improve our model.

After training, the model was evaluated using two standard metrics: Mean Squared Error (MSE) and R² score. These metrics provide insight into the accuracy of predictions and the model's ability to explain the variance in beer ratings.

Finally, the methodology includes an interpretability phase. Feature importance was analyzed to identify the most influential variables, and additional visual tools such as heatmaps and synthetic profile generation were used to explore how combinations of features impact predicted ratings. This step allows not only prediction, but also a deeper understanding of what defines a highly rated beer.

VI. Model Training

The Random Forest model was trained on the prepared dataset using the training subset (*X_train*, *y_train*). During this phase, the model learns to associate beer characteristics with their corresponding ratings by building multiple decision trees.

Each tree is trained on a random subset of the data and features. The final prediction is obtained by averaging the outputs of all trees.

Once the training process is completed, the model is ready to generate predictions on unseen data, which will be used in the evaluation phase.

VII. Model Evaluation

The model was evaluated using the Mean Squared Error (MSE) and the coefficient of determination (R^2) on the test dataset.

The obtained results are summarized as follows:

- MSE = 0.099

- R^2 = 0.455

The MSE value indicates that the average prediction error remains relatively low, suggesting that the model provides reasonably accurate predictions.

The R^2 score of 0.455 indicates that the model explains approximately 45.5% of the variance in beer ratings. This reflects a moderate predictive performance, which is expected given the subjective nature of taste and user rating behavior.

Overall, the model is able to capture meaningful patterns in the data. However, it also highlights that beer appreciation depends on additional factors that are not fully represented in the dataset.

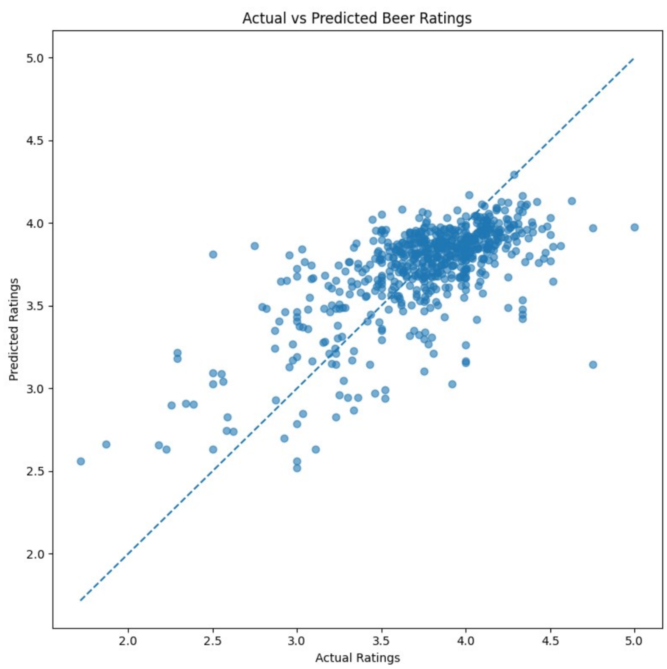

Fig. 3. Actual vs Predicted Beer Ratings

The scatter plot (Figure 3) comparing actual and predicted ratings shows a clear positive correlation, indicating that the model is able to capture the general trend of the data. Most points are concentrated around the diagonal line, which represents perfect predictions. However, some dispersion can be observed, especially for lower ratings, suggesting that the model has more difficulty predicting extreme values.

The error distribution histogram (Figure 4) shows that most prediction errors are centered around zero, which confirms that the model does not exhibit a strong bias. The majority of errors remain relatively small, although a few larger deviations are present. This indicates that while the model performs reasonably well overall, it still struggles with certain cases.

These visualizations confirm the conclusions drawn from the MSE and R^2 metrics: the model provides consistent and moderately accurate predictions, but cannot fully capture the complexity of beer ratings.

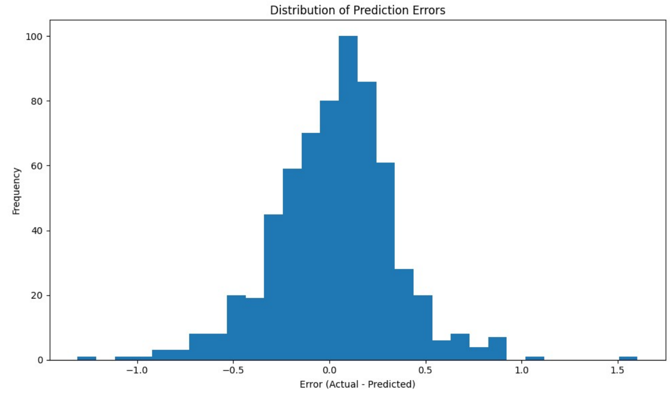

Fig. 4. Error distribution

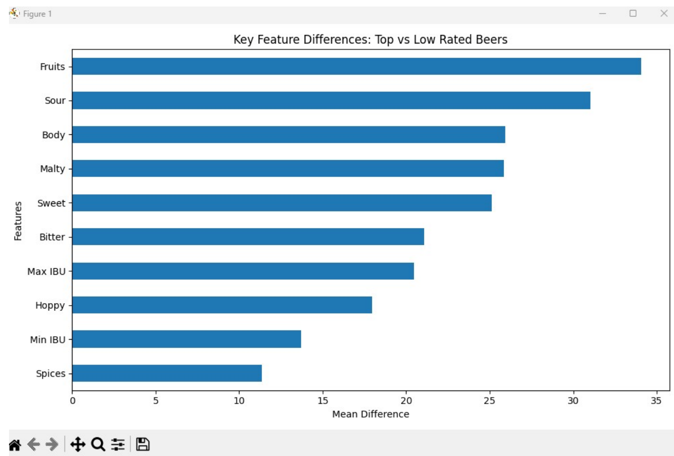

Fig. 5. Key differences between top-rated and low-rated beers

VIII. Results and Interpretation

In this section, we will present and interpret the different graphs produced by our model.

The results obtained from both statistical analysis and machine learning provide valuable insights into the characteristics of highly rated beers.

Fig. 5 highlights the key differences between top-rated and low-rated beers. The most influential features are Fruits, Sour, Body, and Malty, which show significantly higher values in top-rated beers. This suggests that beers with richer and more complex flavor profiles tend to receive better ratings.

Fig. 6 presents the feature importance derived from the Random Forest model. The model identifies Min IBU, Spices, ABV, and Body as the most important predictors. This confirms that both taste-related attributes and bitterness levels play a crucial role in determining beer quality.

Figs. 7–**??** show prediction heatmaps generated by varying two features at a time while keeping others constant. These visualizations reveal how feature interactions influence the predicted rating:

- Fig. 7 (Fruits vs Sour) shows that higher fruitiness combined with low sourness tends to maximize ratings.

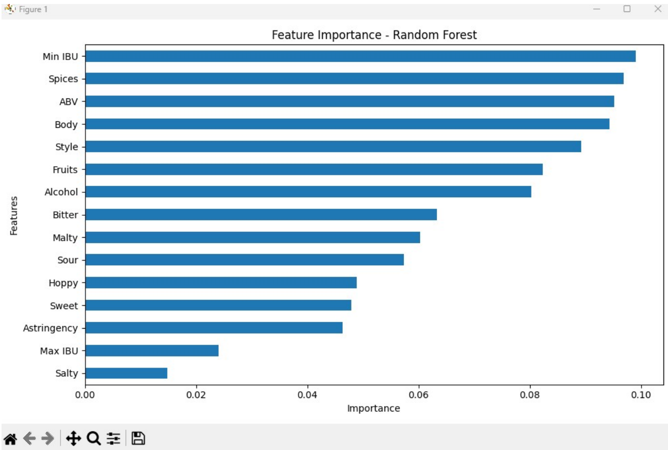

Fig. 6. Feature importance from Random Forest model

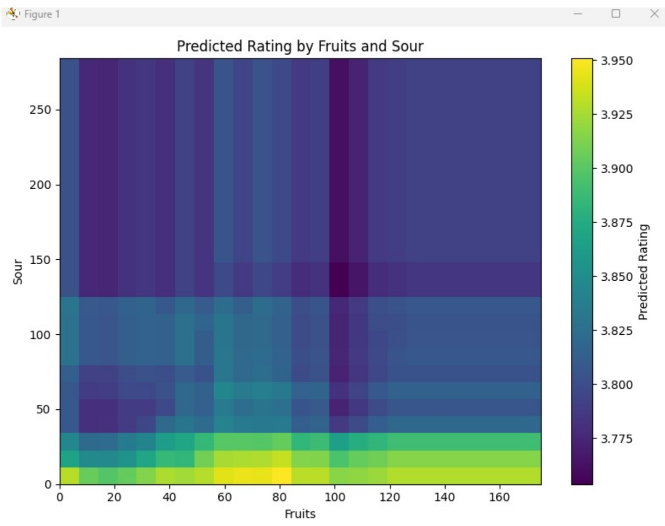

Fig. 7. Prediction heatmap: Fruits vs Sour

- Fig. 8 (Body vs Malty) indicates that fuller-bodied and malty beers generally achieve higher predicted scores.

Finally, Fig. 9 presents the top predicted beer profiles generated by the model. The best combinations consistently involve high Body and Malty values, while keeping Sour close to zero. Interestingly, Fruits does not always need to be high, indicating that texture and maltiness may be more critical than fruitiness alone.

Overall, these results demonstrate that beer quality is influenced by a combination of multiple interacting features rather than a single dominant factor. The model successfully captures these relationships and provides meaningful insights for understanding and optimizing beer profiles.

IX. Discussion

To validate and contextualize our results, we compared our findings with real-world data from the Untappd Top Rated Beers ranking.

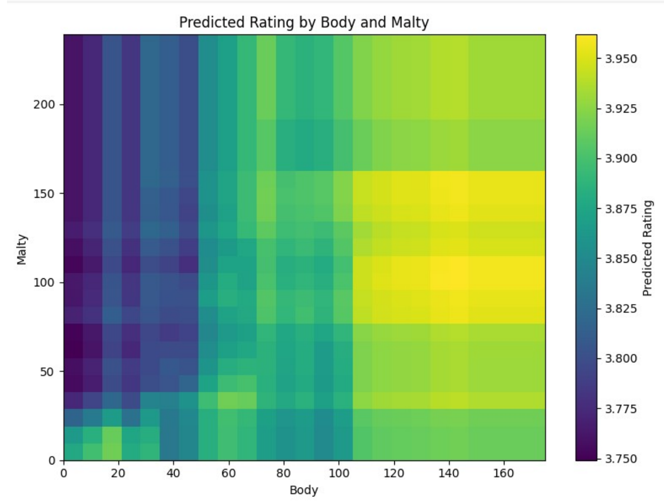

Fig. 8. Prediction heatmap: Body vs Malty

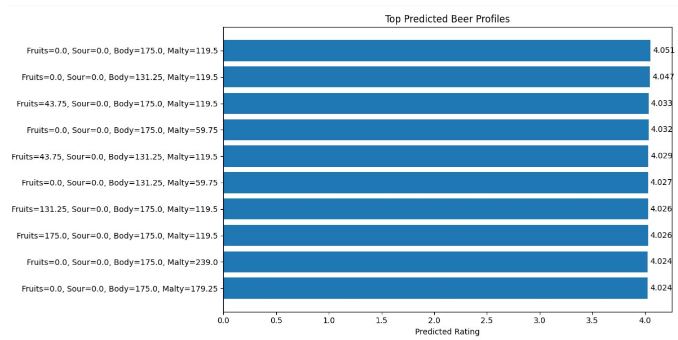

Fig. 9. Top predicted beer profiles

Untappd is a widely used platform where users rate beers based on their tasting experience. The highest-rated beers on this platform are often characterized by strong and complex flavor profiles, including high body, pronounced maltiness, and balanced bitterness.

Our model predictions are consistent with these observations. The top predicted beer profiles generated by our model tend to favor high Body and Malty values, which aligns with the characteristics of highly rated beers on Untappd. Additionally, the model suggests that low sourness contributes positively to ratings, which is also observed in many popular beer styles.

However, a key difference lies in the predicted rating range. While Untappd includes beers rated close to 4.5 or higher, our model predictions remain around 4.0–4.1. This limitation can be explained by the dataset distribution, where extremely high ratings are rare. As a result, the model tends to regress toward the mean and avoids predicting extreme values.

Furthermore, Untappd rankings are influenced by subjective user preferences, trends, and brand reputation, which are not captured in our dataset. Our model focuses only on measurable features such as flavor components and chemical properties.

Overall, this comparison highlights that while our model captures general patterns of beer quality, it does not fully reflect the complexity of real-world perception. Nevertheless, it provides a solid approximation of the factors that contribute to highly rated beers.

X. Conclusion

In this project, we aimed to understand and predict beer ratings based on their intrinsic characteristics using machine learning techniques.

Through exploratory data analysis, we identified key relationships between flavor attributes and overall ratings. Features such as body, maltiness, fruitiness, and sourness were shown to significantly influence beer quality.

We then trained a Random Forest regression model, which achieved moderate performance ($R^2 \approx 0.45$), indicating that a substantial portion of the variance in ratings can be explained by the available features. The model also provides valuable insights into feature importance and interactions between variables.

By generating prediction heatmaps and simulated profiles, we were able to explore how combinations of features impact the predicted rating. This allowed us to approximate optimal beer profiles and better understand what contributes to high-quality beers.

However, the model has several limitations. It does not capture subjective factors such as user preferences, brand reputation, or contextual influences. Additionally, the dataset contains relatively few extremely high ratings, which limits the model's ability to predict values close to 5.

Overall, this project demonstrates that machine learning can effectively model complex relationships in real-world data and provide meaningful insights, even in domains involving subjective evaluation such as beer tasting.

References

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